ENEE459v HW02 : DUE 2/10 (Tuesday) 11:59pm
Please base solutions on extensions of Homework 01's 'newproject_ws'

"Polling" means that a program is reading some status value;
"Blocking" means that the caller waits until the callee finishes its processing.
An "interrupt function" is a void-void function which is triggered by some hardware event external to a program and which runs while the program is paused.

Problem1) Blocking, polling, SysTick_Blinky
Create a CubeIDE project workspace <yourlastname>_HW2P1_ws which produces 1Hz blinks on the green LED attached at PA5 by polling Systick's 'Current Value' register (SYST_CVR) for time information. Ensure that your system clock is running at 80MHz.

Problem2) Non-blocking, polling, SysTick Blinky
Create a CubeIDE project workspace <yourlastname>_HW2P2_ws which uses polling of the blue button's status (pressed, not-pressed) to establish push-ON/push-OFF control over LD1's 1Hz blinking.
(" push-ON/push-OFF " : push once, blinking starts; push again, blinking stops.)

Problem3) Hooking SysTick interrupt for Blinky operation
Create a CubeIDE project workspace <yourlastname>_HW2P3_ws which 'hooks' the SysTick interrupt to create a 1Hz Blinky with push-ON/push-OFF control form the blue button.
The snippet below from Project\Core\Src\stm32l4xx_it.c shows where your function will 'hook' the interrupt:

```
/**
  * @brief This function handles System tick timer.
  */
void SysTick_Handler(void)
{
  /* USER CODE BEGIN SysTick_IRQn 0 */
  <<<your 'hooked' operations go here>>>
  /* USER CODE END SysTick_IRQn 0 */
  HAL_IncTick();
… the rest of the interrupt
```

WHAT TO SUBMIT
Please follow the submission requirements of Homework 01.